DEPLOYING A STATIC WEBSITE ON AWS S3

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INTRODUCTION AND OBJECTIVES

Amazon S3 (Simple Storage Service) is a cloud-based object storage service provided by Amazon Web Services (AWS). It allows users to store and retrieve files over the internet. Amazon S3 also supports Static Website Hosting, enabling users to host websites consisting of HTML, CSS, JavaScript, images, and other static content without using a traditional web server.

In this project, a static website was created and deployed using Amazon S3. The website files were uploaded to an S3 bucket, public access permissions were configured, and static website hosting was enabled. The website was then accessed through the generated S3 website endpoint URL.

• To understand the fundamentals of cloud computing and Amazon Web Services (AWS).

• To study the features and functionalities of Amazon S3 (Simple Storage Service).

• To create and configure an Amazon S3 bucket for storing website files.

• To develop a simple static website using HTML and CSS technologies.

• To upload website content to an Amazon S3 bucket.

• To enable Static Website Hosting for the bucket.

• To configure public access permissions and bucket policies.

• To generate and use the S3 Website Endpoint URL.

• To verify successful website deployment through a web browser.

• To gain practical experience in cloud-based application hosting.

TECHNOLOGIES USED

Technology : Amazon Web Services (AWS)

Purpose : Cloud platform for hosting applications and storage services.

Technology : Amazon S3

Purpose : Object storage service used for static website hosting.

Technology : HTML

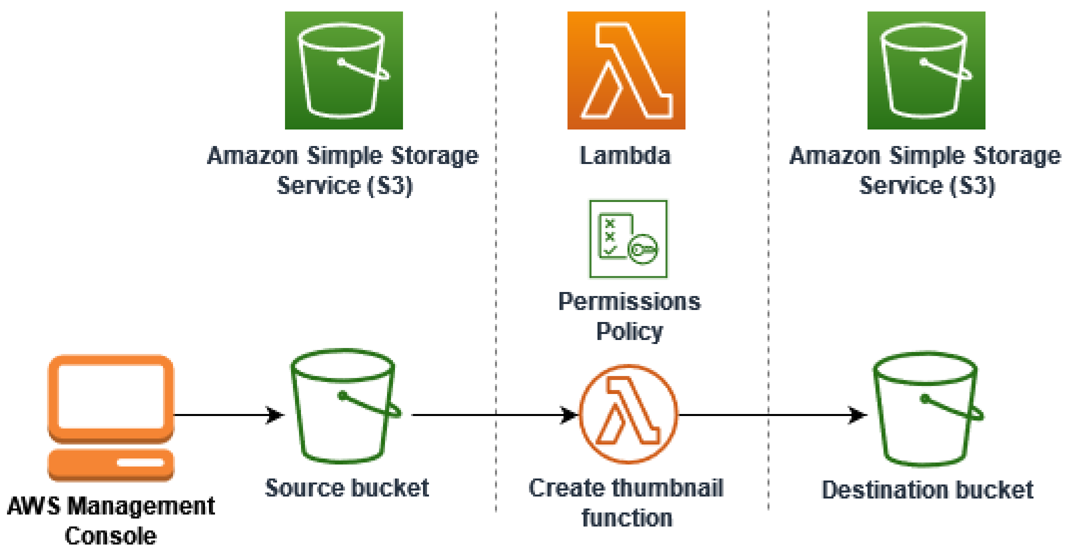
Purpose : Used to create website structure.

Technology : CSS

Purpose : Used to design and style the website.

Technology : AWS Management Console

Purpose : Used to manage AWS resource



STATIC WEBSITE DEPLOYMENT

Resource Details :

- Bucket Name : praveen-s3-website-002
- Region : Asia Pacific (Sydney)
- Hosting Type : Static Website Hosting
- Index Document : index.html
- Error Document : index.html

Deployment Process :

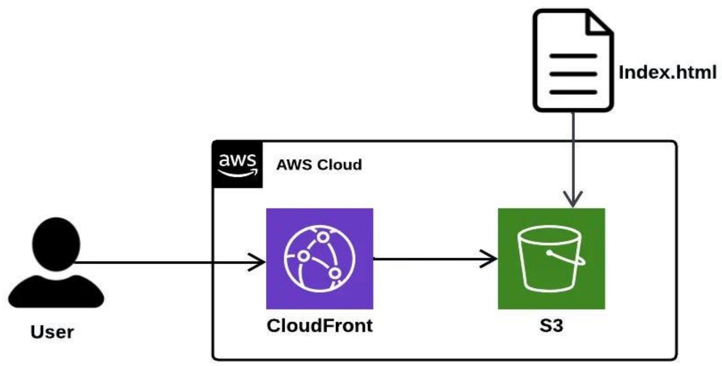
*Logged in to AWS *Management Console.

*Opened Amazon S3 service.

*Created a new S3 bucket.

*Uploaded website files.

*Enabled Static Website Hosting.



IMPLEMENTATION

Figure 1: S3 Bucket Creation

*Amazon S3 bucket "praveen-s3-website-002" was successfully created in the AWS Management Console to host the static website files.

*The bucket serves as the storage location for all website files, including HTML, CSS, images, and other static content. Amazon S3 provides high availability, scalability, durability, and secure storage for hosting static websites.

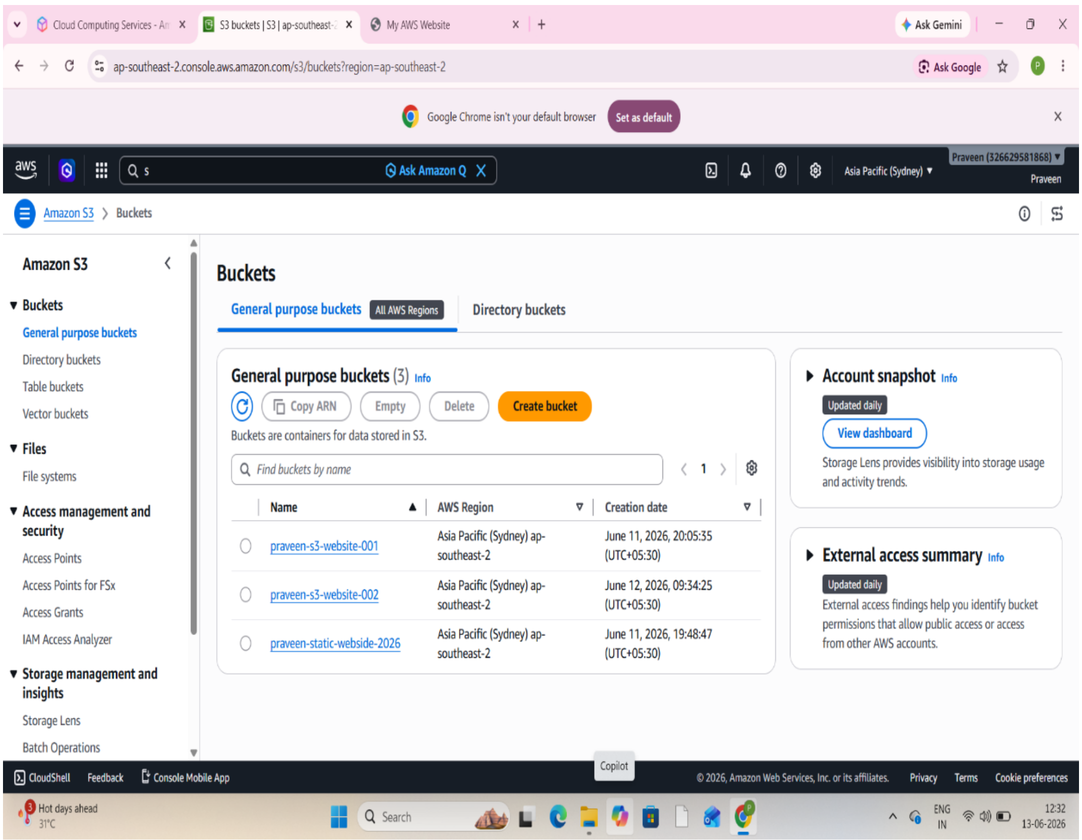


Figure 2: Uploaded Website Files

*The website files index.html and style.css were uploaded successfully to the S3 bucket and stored as website content.

*The website files were successfully uploaded to the Amazon S3 bucket using the AWS Management Console.

*These files were stored as objects within the S3 bucket and became the primary resources used for website hosting.

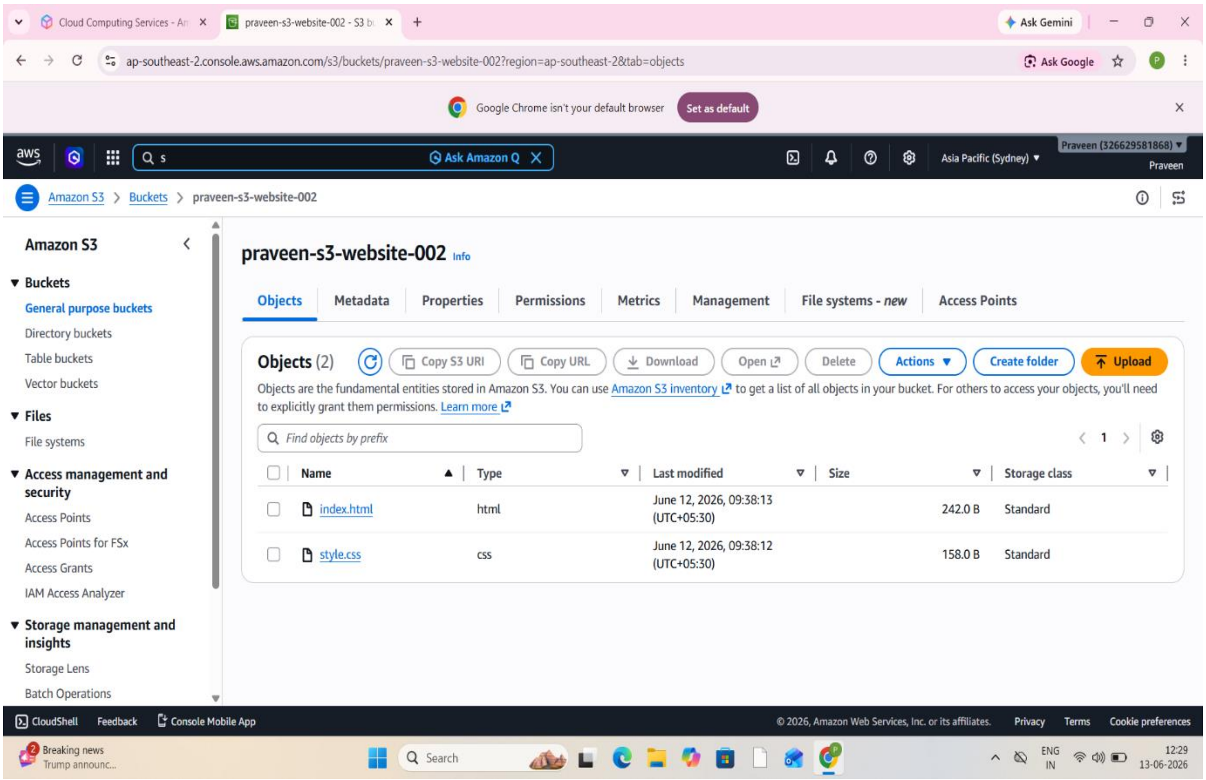


Figure 3: Static Website Hosting Enabled

*Static Website Hosting was enabled in the bucket properties. The S3 website endpoint URL was generated for public access.

*Amazon S3 provides a built-in Static Website Hosting feature that allows users to host static web pages directly from an S3 bucket without requiring a dedicated web server.

*The error document was also configured to handle invalid page requests and improve user experience.

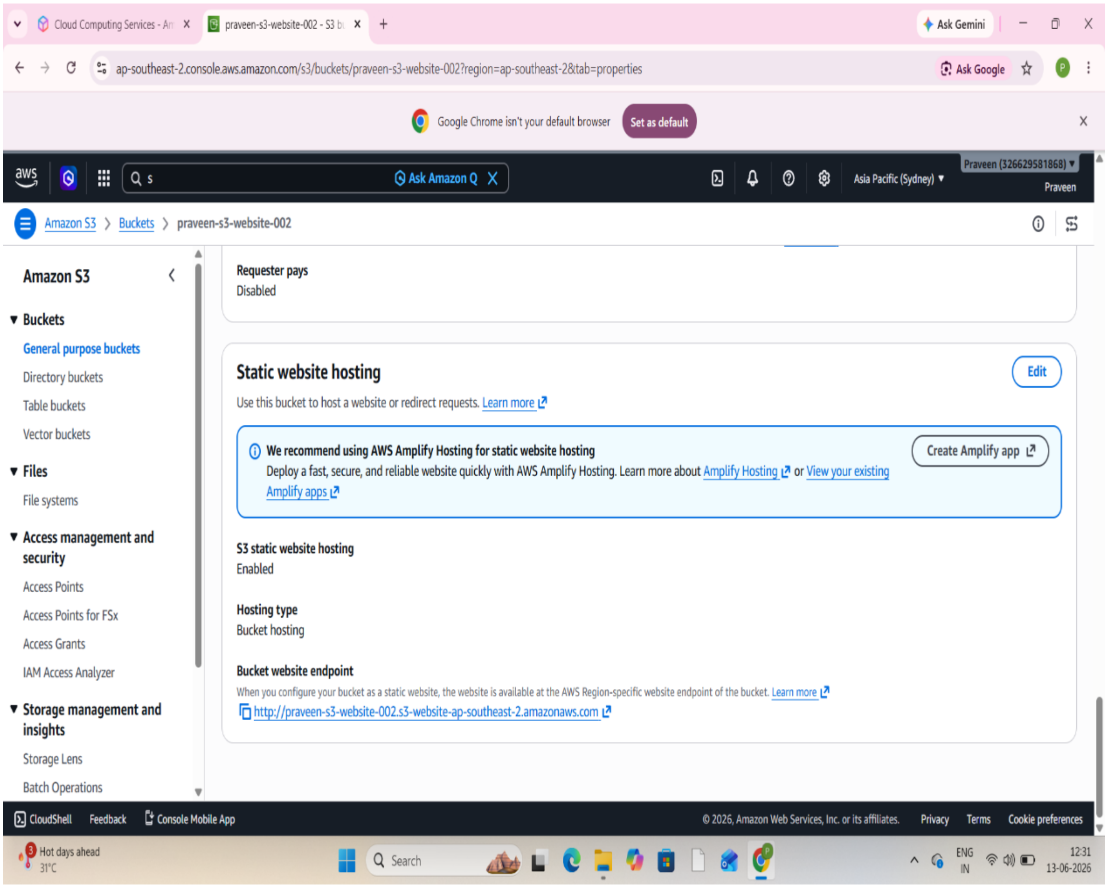


Figure 4: Bucket Policy Configuration

*A Bucket Policy is an AWS Identity and Access Management (IAM) resource policy that defines permissions for accessing objects stored within an Amazon S3 bucket. By default, all S3 buckets and objects are private, meaning that only the bucket owner has access to the stored content. To make the static website publicly accessible, a bucket policy was created and attached to the bucket.

*The policy granted public read access to all objects stored in the bucket. This permission allows users on the internet to view and download website files such as HTML, CSS, JavaScript, and images through the website endpoint URL. Without a properly configured bucket policy, visitors would receive an "Access Denied" error when attempting to access the website.

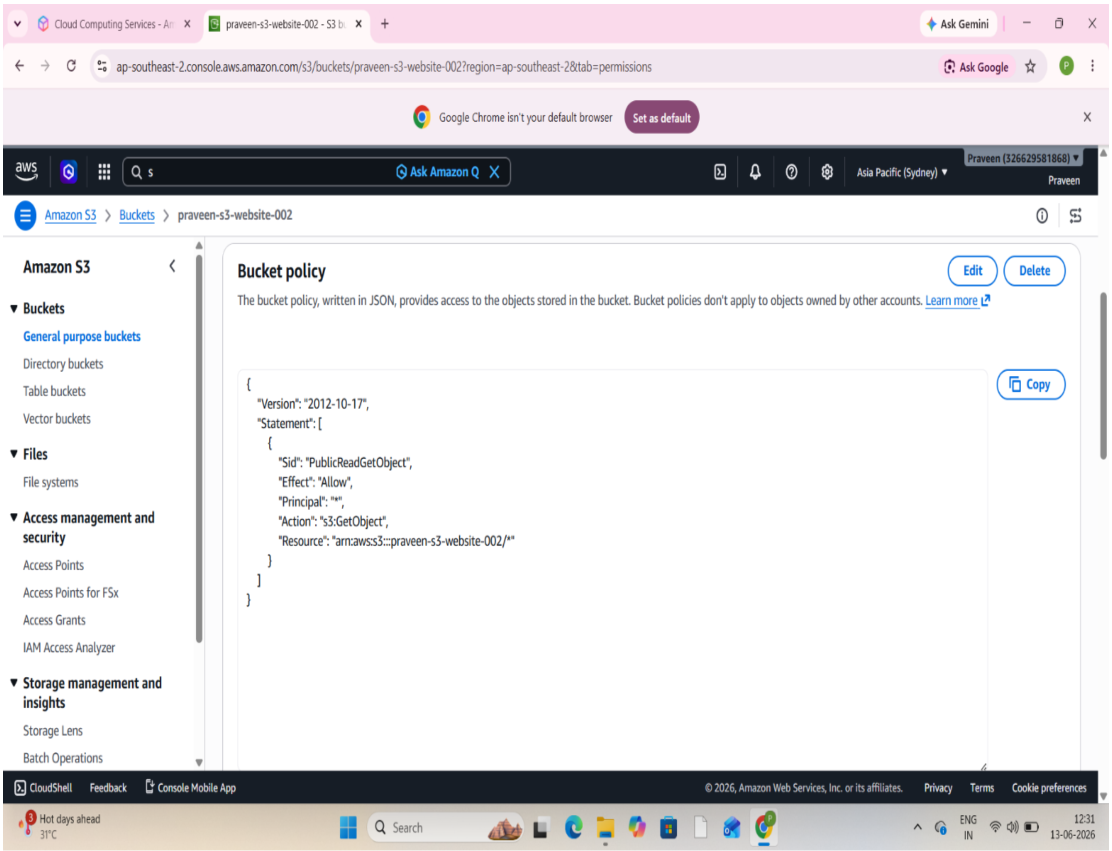
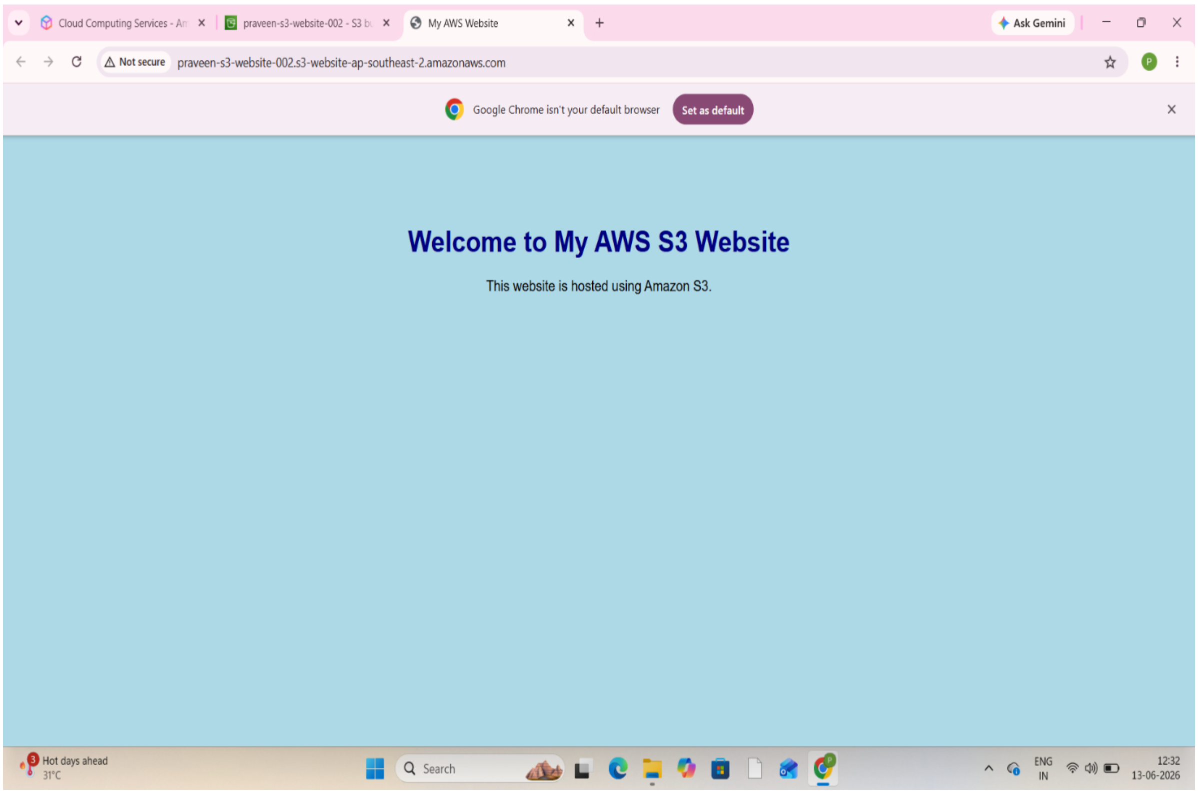


Figure 5: Website Deployment Result

*The static website was successfully deployed using Amazon S3 Static Website Hosting and made publicly accessible through the generated S3 website endpoint URL. After uploading the website files, enabling static website hosting, and configuring the required permissions, the website was tested in a web browser to verify its functionality.

*When the website endpoint URL was accessed, the index.html file was loaded automatically, and the CSS styling was applied correctly. The webpage content was displayed as expected without any errors, confirming that all configurations were completed successfully. This demonstrated the effectiveness of Amazon S3 as a platform for hosting static websites in a cloud environment.

*The successful deployment also verified that the bucket policy, public access settings, and website hosting configurations were functioning correctly. Users can now access the website from anywhere through the internet using the generated URL.



RESULTS AND CONCLUSION

Project Results

The static website was successfully hosted using Amazon S3 Static Website Hosting. All website files were accessible through the generated endpoint URL, and the website displayed correctly in the browser.

Conclusion

This project provided practical experience in cloud computing and website deployment using AWS. Amazon S3 offers a simple, scalable, secure, and cost-effective solution for hosting static websites without requiring dedicated server infrastructure.

PROJECT URL :

Live Website:

http://praveen-s3-website-002.s3-website-ap-southeast-2.amazonaws.com

GitHub Repository:

Praveen-Kumar18/Deploying-a-static-website-on-AWS-S3